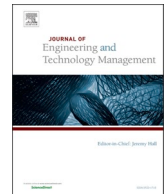




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## Relationship management for recurrent project value creation

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### ABSTRACT

This paper investigates how firms can be the most effective at relationship management to create initial and long-term, recurrent project value. The paper uses qualitative data from in-depth interviews to analyse the project experience of a multinational firm with its strategic partners to better understand inter-firm relationship success. The analysis shows that complementary resources and knowledge sharing are necessary, but not sufficient for value creation from inter-firm relationships, and competitive alignment is introduced to differentiate between the successful and less successful relationships. The research shows the importance of defining competitive alignment during the negotiation process and sustaining it throughout each relationship cycle in order to create long-term value. In addition, practitioners can learn how to exploit formal control mechanisms to incentivize competitive alignment with their external partners.

### 1. Introduction

Despite the growing relationship management literature, researchers have not systematically explored how firms can be the most effective at relationship management to secure value creation during project-based relationships (see Floricel et al. 2023; Kostis et al., 2022; Cao and Lumineau, 2015; Willumsen et al., (2019); Nippa and Reuer, 2019 for reviews). In addition, existing theoretical streams, including transaction cost and the relational view, contradict each other about the effective safeguards against opportunism during project execution. Thus, the aim of this research is to address the gaps between these perspectives while examining the management activities of a company and its long-term relationship management processes including partner selection, contract negotiation, and project execution with multiple partners.

Globalization, external environment dynamism, and rapid innovation in information and communication technologies motivate firms to draw more heavily upon resources and capabilities beyond firm boundaries. Thus, access to partnerships and networks becomes increasingly important for survival, access to innovation and opportunities for growth, and superior value (e.g. Nippa and Reuer, 2019; Dyer et al., 2018; Sydow and Braun, 2018; Martinez et al., 2018; Beamish and Lupton, 2016; Hoang and Rothaermel, 2016). Focus on sharing, cooperation, and interdependence creates space for inter-firm relationships serving as alternative governance mechanisms to those of hierarchical governance, vertical integration, and pure market transactions. These inter-organizational relationships include strategic alliances, joint ventures, joint development agreements, and other collaborative relationships with external partners (Sydow and Braun, 2018; Cao and Lumineau, 2015; Nippa and Reuer, 2019). While we have evidence that these relationships can be valuable for innovation, market-entry, and performance, we still know little about how to be most effective at relationship management, or inter-firm engagement to create partnerships, to create initial and long-term, recurrent project value.

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Transaction cost theory and the relational view shed light on the challenges and opportunities managers face during relationships with external partners. According to transaction cost theory, inter-organizational relationships often involve specialized investments that can lead to co-dependence, opportunism, and unproductive behaviour in the absence of appropriate formal incentives and control mechanisms (Williamson, 1981; Ketokivi and Mahoney, 2020). In contrast, the relational view highlights frequent and high-volume knowledge sharing, relation-specific assets, complementary resources, and informal safeguards as potential sources of inter-firm competitive advantage (Dyer and Singh, 1998; Dyer et al., 2018; Kruse et al., 2022). These relational resources can reduce transaction costs of inter-organizational relationships and create relational value that is greater than the sum of the rents each firm can create independently (Dyer, 1997; Dyer and Singh, 1998; Dyer et al., 2018; Kruse et al., 2022).

The conflicting forces highlighted by these two literature streams raise the question about how inter-organizational relationships could be managed effectively to encourage partners to focus on value creation rather than opportunistic behaviour. Thus, this study's objective is to examine inter-firm relationships with different degrees of success by focusing on key relationship building and management factors. While the subject of inter-firm relationships is very broad, this paper focuses on relationship management, which is inter-firm engagement in order to create partnerships, and in particular it studies the processes of identification and selection of partners, contract negotiation, and project execution. In addition, value creation is a critical and frequently referred to concept in this paper, which we define as the process of turning assets, resources, and capabilities into realized benefits meeting the needs of stakeholders such as partners, customers, investors, and/or employees (see Green and Sergeeva, 2019 for a review). This research focuses specifically on inter-firm relationships and recurrent project value creation. Examples of project value created include project approval, project initiation and completion, and ongoing project benefits, and the value operationalized is not necessarily the same from one project to the next.

This paper's research question is what are the practices for inter-firm relationship management that are most effective for creating initial and long-term recurrent project value creation? Given the complexity of the social processes of relationship management, qualitative data is particularly well equipped to address this question. Thus, using interview and secondary data, the experience of a multinational, waste-to-energy developer with six strategic partners in the UK is analysed to investigate inter-firm relationships. This company fostered relationships with multiple external partners to develop several waste-to-energy projects in the UK over an extended period of time. Thus, the company is uniquely positioned and studying it offers theoretical insights on a relationship management approach for achieving recurrent project value creation with external partners.

## 2. Theoretical background

Transaction cost theory and the relational view help us understand different characteristics of inter-firm relationships. Transaction cost theory posits that monitoring, controlling, and managing transactions result in costs, and that firms should minimize these costs when choosing between internal and market transactions (Williamson 1979, 1986; Carson et al. 2006; Ketokivi and Mahoney, 2020). Transaction costs include finding and vetting partners, negotiating purchases, and monitoring the delivery and quality of goods and services. A firm's ability to assess costs correctly and contractually control the external party influence whether the company integrates additional operations or purchases the services from a supplier.

In contrast, the relational view posits that inter-firm relations can result in combining and co-developing resources that generate competitive advantage when compared to internal operations and market transactions (Kruse et al., 2022; Dyer et al., 2018; Zhe-lyazkov and Gulati, 2016; Dyer and Singh, 1998; Carson et al., 2006). Firms can be successful relying on resources they develop as part of relational networks (Kruse et al., 2022; Arya and Lin, 2007; Dyer and Singh, 1998; Lavie, 2006). These networks offer informal

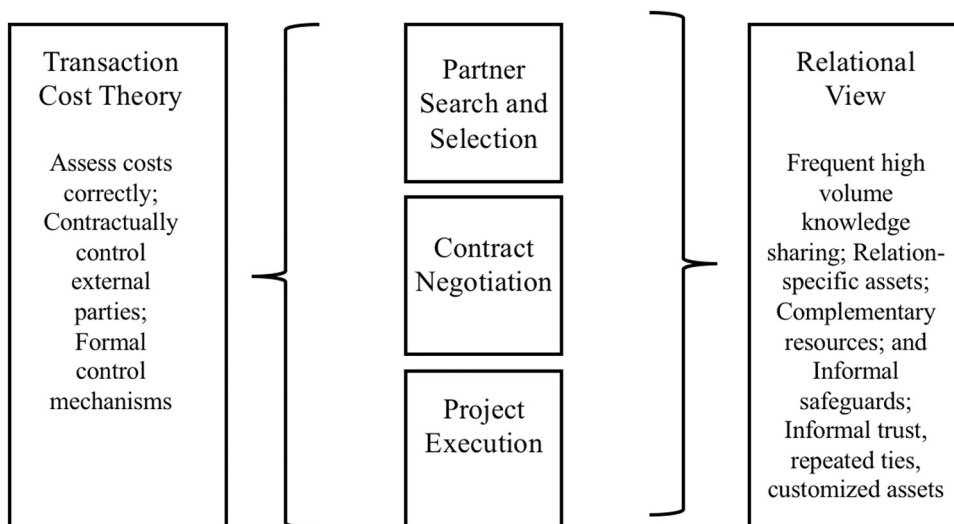


Fig. 1. Conceptual Framework for Relationship Management.

safeguards such as trust and customized assets that lower transaction cost and increase value creation. The relational view describes value creation based on four factors including complementary resources and capabilities, relation-specific assets, knowledge-sharing routines, and effective governance (Dyer and Singh, 1998). Value creation from inter-firm relations is “...the value created in an alliance that is above and beyond the value created in competing arms-length market relationships (Dyer et al. 2018).” Complementary resources and capabilities are why alliances form between entities. Partners build trust by developing relation-specific assets and knowledge-sharing routines with each other through repeated engagement over time. Effective governance, in the form of developed norms and shared experiences, is also necessary for maintaining inter-firm relationships and helping ensure continued abidance to negotiated conditions of the partnership (Kruse et al., 2022; Dyer & Singh, 1998).

Combining insights from these literature streams we match relationship process and content characteristics to summarize factors that influence inter-organizational relationships (Fig. 1 and Table 1). These relationship processes are partner search and selection, contract negotiation, and project execution.

### 2.1. Partner search and selection

Inter-firm relationships start with the identification and selection of a partner. Potential partners become familiar with each other's cultures and personalities, declare intentions and expectations, and identify complementary resources and capabilities (Kruse et al., 2022; Teece, 2014; Ring and Van De Ven (1994)). Thus, partner characteristics play an important role in the selection process.

Three types of partner characteristics with potential to create synergistic value in a relationship are technological, strategic, and relational (Emden et al., 2006). Technological alignment can be achieved with partners who possess resource complementarities, overlapping knowledge bases, and technical abilities (Emden et al., 2006). Complementary resources that offer competitive advantage can represent valuable assets (Kruse et al., 2022; Rauniar et al., 2019; Teece, 1986; Teece, 2014; Dyer and Singh, 1998; Aggarwal et al., 2011). They can be defined as different parts of an ecosystem brought together by different parties to develop and commercialize new products or services (Teece, 1986). Distinctions are drawn between generic or unmodified resources and specialized or tailored resources. In addition to identifying and evaluating partners with complementary resources, the ability to leverage these resources of external partners is crucial to the success of inter-firm relationships (Kruse et al., 2022; Luo, 1997, 2007; Hill and Hellriegel, 1994; Dyer and Singh, 1998). Strategic alignment refers to partners' collaboration capabilities, similar motivation, and goal correspondence (Emden et al., 2006; Shi et al., 2012). The relational alignment encompasses compatible cultures, propensity to change, and long-term orientation of partners important for the sustainability of relationships (Rauniar, et al., 2019; Emden et al., 2006). Similarities in organizational culture and values can help partners understand each other better and improve relationship productivity (Boddy et al., 2000).

Finally, in addition to these alignments, firms pay attention to partners' reputation as another selection criterion (Roy and Oliver, 2009; Saxton, 1997; Aggarwal and Hsu, 2009). Firms are motivated to create relationships with reputable partners because a good reputation is often seen as a sign of expertise, legitimacy, and fairness (Carson et al., 2006).

### 2.2. Contract negotiation

A second element of the relationship development process involves contract negotiation. Potential partners go through a bargaining process about input and output conditions of the relationship such as division of tasks and responsibilities, allocation of resources, and distribution of outcomes (Suprpto et al., 2016; Reuer et al., 2013; Ring and Van De Ven (1994)). The alliance literature and transaction cost theory refer to contractual arrangements as a tool to ensure task completion and distribution of returns (Teece, 2014; Ariño and de la Torre, 1998; Pisano and Teece, 2007; Adegbesan and Higgins (2011); Yan and Gray, 1994)). In addition, the protection of the

**Table 1**  
Relationship Process and Content.

| Relationship Process         | Relationship Content                                                                                                                                                                                              | Selected References                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner Search and Selection | <ul style="list-style-type: none"> <li>– Complementary resources</li> <li>– Similar motivation and goal correspondence</li> <li>– Compatible cultures</li> <li>– Reputation</li> <li>– Trustworthiness</li> </ul> | Kruse et al., 2022; Geringer, 1991; Teece, 1986; Teece, 2014; Ring and Van De Ven (1994); Dyer and Singh, 1998; Emden et al., 2006; Aggarwal et al., 2011; Contractor and Reuer, 2014; Luo, 1997; Luo, 2007; Roy and Oliver, 2009; Shi et al., 2012 |
| Contract Negotiation         | <ul style="list-style-type: none"> <li>– Task completion</li> <li>– Distribution of returns</li> <li>– Intellectual property rights</li> </ul>                                                                    | Ring and Van De Ven (1994); Ariño and de la Torre, 1998; Pisano and Teece, 2007; Adegbesan and Higgins (2011); Reuer et al., 2013; Yan and Gray, 1994; Suprpto et al., 2016; Kruse et al., 2022                                                     |
| Project Execution            | <ul style="list-style-type: none"> <li>– Knowledge management</li> <li>– Opportunistic behaviour</li> <li>– Shared-decision making</li> </ul>                                                                     | Saxton, 1997; Luo, 2005; Williamson, 1985; Grant, 1996; Luo, 2008; Dyer, 1997; Roy, 2012; Carson et al., 2006; Lyles and Salk (1996); Athaide and Klink, 2009; Verdecho et al., 2012; Stjerne et al., (2019)                                        |

intellectual property is a central issue discussed during the negotiation of contractual agreements (Teece, 1986). Most innovative firms view intellectual property, in the form of patents, trademarks, copyrights, and licenses, as a valuable resource contributing to their competitive advantage. Thus, firms rely on contractual agreements to protect against infringements of intellectual property created prior to the relationship and facilitate knowledge sharing. The negotiation process also includes discussions about the allocation of know-how developed during the relationship to ensure adequate value distribution (Adegbesan and Higgins (2011)).

### 2.3. Project execution

A third element of the relationship cycle is project execution, during which the responsibilities and activities defined during the negotiation process and formalized in the contractual agreements are set in motion (Kruse et al., 2022; Roy, 2012; Ring and Van De Ven (1994)). Collaboration and personal interactions facilitate the execution of joint responsibilities and activities. Thus, the interrelated processes of communication, project coordination, and knowledge management become essential (Kruse et al., 2022; Luo, 2005; 2008; Lyles and Sulk, 1996; Grant, 1996; Athaide and Klink, 2009; Verdecho et al., 2012). Theoretical streams such as transaction cost and the relational view agree that greater frequency and volume of knowledge exchange increase partner commitment and improve cooperation (Dyer, 1997; Williamson, 1985). Shared-decision making is another factor that positively affects project execution (Saxton, 1997). In contrast, opportunistic behaviour characterized by unnecessary bargaining, unfulfilled obligations, and withholding valuable information is often seen as a detriment of project productivity (Williamson, 1985; Carson et al., 2006; Ketokivi and Mahoney, 2020).

Once the first relationship cycle or project is completed, firms may either terminate or extend the relationship through engagement in another project. The potential benefits of repeated and more in-depth engagement (building relational capital) are built trust, established processes, shared risk-taking and financing, shorter negotiation time, fewer contractual hurdles, and innovation (Kruse et al., 2022; Buvik and Rolfsen, 2015; Ring and Van De Ven (1994); Dyer, 1997). The challenges include changed alignments over time, loss of competitive edge, and opportunistic behaviour by one party. In the case of preference for a continuous relationship, uncertainty around contextual changes and people's responses towards these changes create demand for re-assessment of the relationship conditions (Buvik and Rolfsen, 2015; Ring and Van De Ven (1994); Boddy et al., 2000; Carson et al., 2006). As a result, the modification of partner behaviour and renegotiation of contractual terms can occur (Ring and Van De Ven (1994); Ariño and de la Torre, 1998). Prior affiliation with a partner can motivate re-engagement and increase the satisfaction of partners participating in recurrent projects (Saxton, 1997). The institutional ties and inertia developed from prior partnering experiences are likely to increase the propensity to re-engage with the same partner, but they are not sufficient to guarantee value creation and interest alignment during the consequent engagements (Kruse et al., 2022; Dyer et al., 2018).

Despite our growing understanding about relationship management, researchers have not systematically explored how characteristics within each process element contribute to value creation and relationship success. Furthermore, we lack understanding about how firms can be most effective at relationship management to secure value creation beyond the initial engagements with the same partner to sustain long-term relationships (Dyer et al. 2018). Thus, the research question of this paper is what are the practices for inter-firm relationship management that are most effective for creating initial and long-term recurrent project value creation?

## 3. Data collection methods and analysis

### 3.1. Data collection methods

The research question for this paper addresses complex social processes of relationship management; thus, qualitative data was used to explore how these processes occur and develop new theoretical insights. To investigate our research question, we used a nested interview approach (Mueller, 2019) to analyse Covanta's experience with six strategic partners during the development of multiple waste-to-energy projects. We chose Covanta using theoretical sampling (Lune and Berg, 2018; Eisenhardt and Graebner, 2007). First, we narrowed down our selection to a few companies by focusing on only renewable energy companies. Within that group we looked at firms engaged in multiple inter-firm relationships. Finally, we selected Covanta because the company is well positioned to offer theoretical insight into the relationship management approaches due to its experience with multiple partners. This enabled us to analyse multiple relationships within one organization. We gained insight into the opportunities and challenges Covanta faced during each relationship from multiple perspectives, including views from the senior managers within Covanta and from partner organizations. Thus, the research rests on six inter-firm relationships involving at least one project within a single context using a nested interview approach (Lune and Berg, 2018). Each project, rather than a relationship, is used as the unit of analysis. A project can be seen as an event or a critical incident 'when parties engage in actions related to the development of their relationship' facilitating both organizational and individual levels of analysis (Ring and Van De Ven (1994)).

The data were gathered through a series of face-to-face and telephone interviews (12 total interviews, each lasting one to two hours) with the focal firm and its six partners. The interview guide contained semi-structured, open-ended and some ad hoc questions. Semi-structured and open-ended questions were derived from previous studies and the academic literature. The interview questions enabled the interviewees to speak freely on the topic of interest within a provided structure. The interview guide was pretested with businesspeople and academics to ensure clarity of the questions. All the relationships entailed cross-border engagement, and all the people interviewed had worked in multiple countries. None of the interviewees mentioned cultural challenges in the relationships studied.

### 3.2. Data analysis

First, the interview material was analysed through an iterative process. We relied on several rounds of theory and data comparison to facilitate theory building about complex social processes and management capabilities overlooked by prior research (Flick, 2009; Lune and Berg, 2018). We organized and analysed the interview transcripts and secondary data with the help of a content analysis software program—NVivo (Version12). The materials were coded to interpret the text, capture key elements, and develop theoretical perspectives. The software allowed for identification of common themes, frequent terms, connection, and visualization of key concepts for theory development, and comparisons between researchers. The initial analysis was presented to the focal company to identify knowledge gaps and obtain feedback. Second, the interviews with senior managers from multiple organizations offered a portrayal of different perspectives and a broader understanding of complex interactions. Cross-referencing of data from multiple informants allowed us to avoid the problem of single informant bias and convergent retrospective sense making. As a result, greater credibility and external validity of the data and more robust and testable theory was achieved (Lune and Berg, 2018; Eisenhardt and Graebner, 2007). Third, the data was organized into a set of relationships aiming for pattern recognition and logic replication instead of focusing on the rich description of a social setting of a single case. An array of relationships offered independent support for a specific proposition, thus improving the quality, and contributing to generalizability of the emergent theory (Lune and Berg, 2018). Final conclusions were made when constructs matched the empirical evidence of several relationships in order to meet the criteria of rigorous science. Finally, direct quotes were used to provide objectivity, support the analysis, and demonstrate the interviewees' perceptions of the management practices in their own terms.

## 4. Empirical context

### 4.1. Waste-to-energy and covanta

The empirical context for this study is the waste-to-energy (renewable energy) industry in the United Kingdom and incineration projects. This context is particularly suitable for our study for two reasons. First, the industry's emerging and growing nature along with legislative changes requires both innovation and inter-firm collaboration for development, construction, and operation of waste-to-energy projects. Thus, this rarely examined context offers a dynamic and complex environment facilitating industrial and firm changes and offering insights to the academic community. Second, the long-term relationships involving multiple projects with the same partners and creating conditions for recurrent value creation are common due to geographical dispersion and the individual nature of the projects. Therefore, the lessons learned from this dynamic and complex context where partnering is critical for success offer insight to other industries facing uncertain environments where the strength of one's teams can offer competitive advantage and value creation.

Covanta is one of the world's largest owners and operators of waste-to-energy (energy-from-waste or EfW) plants with \$1.7 billion in revenue (2018) and 3600 employees worldwide (Covanta, 2018). While most of its plants have been developed in North America, the company has been focusing on the development of incineration plants in Europe and Asia since 2005. In 2005, Covanta formed a UK subsidiary to focus on the development of new EfW projects in the UK and Ireland.

The next section presents Covanta's relationships with its key partners, including identification and selection of partners, contract negotiation, and project execution, that illuminate the nature of successful engagements and recurrent value creation (see Table 2 for summary details). Covanta and its six partners have international operations, making all of the relationships multinational. The names of the partners have been anonymized for confidentiality purposes.

**Table 2**  
Covanta's Partners.

| Partner name               | # of Employ | Annual Revenue (£ million 2018) | Contract type                                                                                                                       | Length of the relationship |
|----------------------------|-------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ZZ (technology provider)   | 200         | N/A                             | Country specific licensing agreement                                                                                                | 30 years (terminated)      |
| YY (technology provider)   | 10          | 3                               | Joint licensing and development agreement                                                                                           | 3 years (terminated)       |
| XX (technology provider)   | 300         | 110                             | Project by project exclusivity                                                                                                      | 12 years (on-going)        |
| WW (utility/site provider) | 200         | 100                             | JV 50/50 Focal and WW Water Services                                                                                                | 2 years (terminated)       |
| VV (site provider)         | 6200        | 15                              | JV 50/50 UU and XX                                                                                                                  | 11 years (on-going)        |
| UU (construction)          | 50,000      | 11,000                          | JV 50/50 Focal and VV for infrastructure, royalty<br>JV between XX and UU, procurement and construction contractor to focal company | 9 years (on-going)         |



#### 4.2. ZZ (Technology provider)

The emergence of the waste disposal industry in the United States was spurred by change in the landfill regulations in the early 1980 s. This was facilitated by strong growth in municipal solid waste and a decrease in the availability of landfill space (Bator and Stansell, 2005). Covanta looked for a partner with waste-to-energy technology to enter the waste treatment business in the United States and decided to enter into a strategic partnership with technology provider ZZ, due to its experience in the development and commercialization of a waste recovery technology in the European markets.

ZZ developed and commercialized systems for waste processing. This allowed Covanta to benefit from ZZ's specialized complementary resources such as innovative technology as well as operational know-how from previous operations, which helped Covanta enter the waste-to-energy market in the United States. The innovation capabilities and know-how of ZZ motivated the formation of an exclusive 20-year arrangement between these two companies. The agreement included exclusive access to ZZ's technology, thus, limiting anyone else from installing ZZ's technology in the USA and Canada.

As a new waste-to-energy operator, Covanta depended heavily on the knowledge ZZ accumulated from previous operations in the European waste energy market. The partnership between the two companies allowed Covanta to gain knowledge about commercial systems for processing and composting waste. In addition, ZZ's technological expertise and assistance facilitated the identification of potential operational problems. This insight was very useful and enabled Covanta to apply this knowledge to the US market. Covanta VP of R&D described, 'As a new company starting our operations [we] depended very heavily on ZZ because of their technology and know-how and experience from previous operations many years before that.' However, overtime Covanta became less dependent on ZZ's knowledge. Having built several plants, Covanta shifted its growth strategy and started to rely increasingly on acquisitions to sustain its growth. In addition to deploying ZZ's technology, it also engaged with other technology providers diversifying its technology portfolio and operational expertise. Thus, Covanta became the main knowledge provider over time offering ZZ its operational know-how gained from plant operation and maintenance and insights about other technologies. As Covanta UK, CEO stated, 'We continued the collaboration very strongly but after a while it became very much 100% Covanta and nothing given from the other side. I think we'd just overtaken them. I don't think they had anything to add really.'

The exclusive access to ZZ's technology in the US and Canada enabled Covanta not only to integrate the technology into their incineration plants but to also co-develop some elements of ZZ's technology. Within geographical boundaries, the two companies were competitively aligned, as both parties fulfilled their obligations and contributed to each other's success. This created interdependence between the two partners and led to an extension of the initial formal arrangement for another 10 years. This competitive alignment is described by Covanta UK, CEO when he stated, '...both companies were in agreement on which market opportunities to pursue together and how to be profitable.' In addition, both companies were specific about the terms of the technology agreement and how they would share the profits, and the agreement moderated the relationship by signalling shared commitment to the relationship. As a result of the relationship, Covanta developed 25 plants in the US based on ZZ's technology. However, Covanta decided to expand globally by entering the Chinese, Italian, and UK markets. Covanta soon realized that expansion to other geographies challenged their prior partnerships, as they no longer offered competitive edge in new markets. As a result of misaligned competitiveness, Covanta was eventually forced to terminate the agreement with ZZ in the United States. This shows causal links between competitive alignment and value creation, initially in the form of project development in the US for both parties, but later the lack of alignment and lack of access to new markets with Covanta by ZZ.

#### 4.3. YY (Technology provider)

Starting in 2008, Covanta decided to proactively explore other waste-to-energy technologies that could potentially prove to be more effective than the conventional combustion technology used in their existing plants. As a result, Covanta partnered with a small start-up company called YY, specializing in the development of a new and efficient plasma-enhanced gasifier. Interested in a new technology, Covanta formed an exclusive partnership with YY with an aim to assist it with further development and commercialization of the plasma technology by providing capital and defining parameters for the operational system. According to YY's president, 'They are essentially using us as an extension of their R&D. We are doing the research for them and the development of technology or components because they don't have the in-house resources to do that.' YY offered specialized complementary resources tailored to Covanta's operational needs in return for Covanta's generic complementary assets such as capital and existing know-how. In the beginning this alignment was mutually beneficial for both parties, as described by YY's president as '...a good fit in order to exploit market opportunities.' The technology agreement and the capital and know-how moderated the relationship by signalling shared commitment to the relationship.

Covanta gave YY freedom to design an innovative waste-to-energy process suitable for Covanta's operational needs. In the YY presidents' own words, '[Covanta] was not watching over our shoulders because [they] had trust in our capabilities to do it. That kind of trust helps us in moving forward.' However, by granting operational freedom to YY, Covanta failed to keep a close eye on the scope of the project. It allowed a small company lacking operational experience to be involved in the detailed design of a large plant instead of focusing just on the development of a new technology. This resulted in the development of an expensive waste-to-energy process offering low return on investment. As a result of competitive misalignment and lack of formal safeguards, after three years of joint development Covanta decided not to continue with the relationship because it felt that the market potential for new technology was not high enough, as offered by other technological alternatives. This highlights the importance of partners having checks and balances (safeguards) in place during the evolution of the relationship, and it shows the importance of management practices for assessing how to succeed at partner search and selection and competitive alignment throughout the stages of the relationship process.

#### 4.4. XX (Technology provider)

When Covanta entered the UK market in 2005 it looked for a new technology provider due to its inability to access ZZ's technology. Consequently, a licensing agreement between Covanta and XX was initiated in 2007. XX's innovative technology for thermal waste treatment plants represented a specialized complementary resource to Covanta's project development and operational capabilities. XX designed and developed their own technology and customized its technology to address the complexity and versatility of different projects. Innovation was an integral part of XX's activities, as they offered tailored technological solutions to fit individual project requirements.

The relationship between Covanta and XX had a balanced exchange of knowledge, with input from both sides. Initially, this relationship only included one project, but Covanta decided to extend the relationship to two other projects aiming to have a long-term relationship. However, having established that they are 'reliable', Covanta also wanted to ensure that as a preferred partner XX would remain competitive in the long term. In the Covanta UK CEO's own words, 'When we're looking at designs and integration of them into a system, we want it to be as good and efficient and reliable as possible, but we also want it to be at a price we believe will make us competitive. So, there's always that tension and I would say that's the biggest problem.' Prior affiliation, competitiveness, and development of a joint system motivated Covanta to sustain the relationship with XX as the two grew together through diligent communication, sharing, and co-development. This example shows how both partners were able to assess each other's complementary resources, shared knowledge, and competitive alignment as their relationship matured through the stages of development as well as from one project to the next. Each new project represented value creation for both parties supporting a causal link between relationship management and value creation.

#### 4.5. WW (Utility/Site provider)

Covanta bid for a 25-year waste treatment tender with the Bradford Council in the United Kingdom. The relationship with WW did not actually result in a project, but there was an extensive, two-year vetting and engagement period that resembled a project in many ways and makes this a worthy example to include as part of this study. As part of the bid, the company would build and operate a 260,000 ton per year energy-from-waste facility. To win the bid, it initiated a procurement partnership with the utility/site provider, WW, to secure the site for plant construction and off-take the energy generated by the plant. Covanta chose to partner with WW, as it owned a site appropriate for the construction of the plant and was willing to purchase energy generated by the plant, thus, offered generic complementary resources. The knowledge sharing between Covanta and WW was regular and effective. According to WW's senior executive, Covanta proved to be a reliable and transparent partner. A common goal, face-to-face discussions, interpersonal relationships, and timely conflict resolution were very important for knowledge sharing. Through this openness WW was able to gain insights into Covanta's process production and become familiar with their corporate culture and decision-making process.

However, the relationship lacked competitive alignment regarding financial expectations. Despite of having a better understanding about what it takes to win a project, Covanta had to comply with the capital requirements of its partner, who was less familiar with the project development requirements of waste-to-energy projects of such large magnitude. The 50/50 joint venture arrangement and capital rationing of the partner limited the extent to which Covanta could lower its bid price to local authorities to win the project. As a result of competitive misalignment related to capital rationing, the Covanta/WW consortium was outbid by another waste company that offered more favourable conditions to local authorities. Both parties did not expect further cooperation in the foreseeable future. Covanta UK CEO described this relationship as a '...mismatch of financial and strategic positioning.' This example shows the causal link between assessment of partners (evaluation of resources, knowledge, and alignment) and (lack of) value creation in the case of the failed project bid.

#### 4.6. VV (Site provider)

VV was a subsidiary of a property developer focused on the development of new infrastructure projects in the waste sector. VV initiated a tender for the development of a 95 MW waste-to-energy plant. Prior to that VV had obtained planning permission from the local authorities to build an eco-park with waste treatment facilities on its property. As a property developer, VV did not engage in the technology development and plant operations. Thus, the company actively marketed the opportunity to Covanta to build and operate a waste-to-energy facility.

The two companies had no problems with knowledge sharing due to good interpersonal relationships between senior management of the two companies. However, again due to the nature of the relationship Covanta was a key supplier of knowledge in this relationship. Covanta emphasized that they offered a timely and 'regular' supply of information to ensure that VV had 'enough information' to respond to their stakeholders.

VV's aspiration for an eco-park was to develop a cluster of different waste treatment facilities. Thus, the company initially looked for a partner that could not only build and operate an energy-from-waste facility, but also would attract other waste operators and help develop the entire eco-park. Therefore, they looked for a partner with similar vision and competitive alignment. This alignment was in the form of the markets to pursue (positioning) and how each party would benefit (value sharing). This co-dependence represents a formal safeguard ensuring that the partners are financially bound leading to long-term commitment. In terms of future projects, the Director of VV stated, 'Going forward I see no reasons that we can't have a long-term relationship because that would be beneficial on both sides.' The two partners shared complimentary resources and competitive alignment, and they reviewed their objectives periodically to maintain their collaboration. The value creation for Covanta and VV is in the form of project development, which is only

possible if both parties assess resources, knowledge, and alignment of each other during the stages of the relationship process.

#### 4.7. UU (Construction)

The development of waste-to-energy plants includes infrastructure development such as the design and construction of roadways and buildings. Therefore, in addition to a technology provider, site provider, and energy off-taker, Covanta needed to form a partnership with a procurement and construction contractor in the United Kingdom. Covanta decided to partner with UU because of their strong reputation, extensive experience with delivering large-scale civil engineering projects, and innovation capabilities.

In terms of knowledge sharing, there has been a good and open knowledge flow, though the knowledge was supplied mostly by UU to Covanta, and Covanta able to benefit from UU's engineering capabilities in the initial project. This motivated them to consider UU as a preferred partner for future projects. Covanta believed that competitive alignment already existed between the two partners, however, to ensure competitive alignment with UU in the long-term, Covanta proposed to UU a financial union in the form of a requirement for co-investment for subsequent projects. Both parties established clear visions of sharing risk and how to share value creation. In addition, the financial union moderated the relationship by signalling shared commitment to the relationship. The CEO of Covanta UK described, 'We could do a multi-project deal here, which is helpful for all of us, you can plan ahead, but in return I want you to give me the proposal for how much you are prepared to invest upfront in the development of these projects.'

### 5. Analysis and findings

This paper uses the relationships between Covanta and its partners to assess identification and selection of partners, contract negotiation, and project execution to illustrate factors affecting inter-firm relationships summarized in Table 3. The main contribution is an understanding of the roles of complementary resources, knowledge sharing, competitive alignment for value creation, and safeguards (informal and formal mechanisms of control) for competitive alignment. The following are propositions from the analysis of Covanta's relationships that add to our understanding of relationship management.

#### 5.1. Complementary resources and knowledge sharing

Prior research suggests that in relationships where partners offer complementary resources, they also offer inter-organizational competitive advantage and create relational value (Kruse et al., 2022; Rauniar, et al., 2019; Boddy et al., 1998; Emden, et al., 2006; Aggarwal et al., 2011; Contractor and Reuer, 2014; Luo, 2007; Roy and Oliver, 2009; Shi et al., 2012). The evidence from this paper's study shows that complementary resources and knowledge sharing existed in all six relationships. The nature of complementary resources varied from generic to specialized assets including innovative and customized technologies, site provision, energy off-take, and procurement services. The volume and frequency of knowledge exchange varied throughout the cycle of the relationship, and one party was a dominant knowledge provider in most relationships. Generally, the need for knowledge about the products,

**Table 3**  
Relationship Internal Factors and Value Creation.

| Company | Comple-mentary Resources                           | Knowledge Sharing                                                                                                                  | Competitive Alignment                                                                            | Safeguards for Competitive Alignment                                                                                                   | Value Creation (# of Projects)                   |
|---------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| YY      | Innovative technology                              | Regular knowledge exchange, driven by both partners, high reliance on formal agreement prior to knowledge exchange                 | Not aligned due to different expectations regarding technology commercialization cost and timing | Informal safeguards (mutual trust)<br>Lack of formal safeguards (contract limited to IP, but not the project scope, cost and ROI)      | Limited (1)                                      |
| WW      | Site and energy off-take                           | Regular knowledge exchange, driven by one partner                                                                                  | Not aligned due to different capital rationing                                                   | Informal safeguards (trust)<br>Lack of formal safeguards (limited to one project)                                                      | Limited (1)                                      |
| ZZ      | Innovative technologies                            | Regular knowledge exchange, driven by both partners in the beginning, and by one partner after first 5 years                       | Not aligned after global expansion                                                               | Lack of formal safeguards (inability to capture value from co-developed technology, and access to technology outside of North America) | Limited after (25 in USA, none in UK)            |
| XX      | Customized and innovative technology               | Regular knowledge exchange, driven by both partners, amount fluctuated throughout the lifecycle                                    | Aligned to improve quality while minimizing costs                                                | Informal safeguards (reliability, quality)<br>Formal safeguards (competitive price)                                                    | Strong, with potential for recurrent value (2 +) |
| VV      | Access to the site, planning consent, and eco-park | Regular knowledge exchange, driven by one partner, information sharing fluctuated over time                                        | Aligned to develop the eco-park and a waste-to energy plant                                      | Formal safeguards (royalty payment tied to waste quantity for 25 years, and two projects bundled together)                             | Strong, with potential for recurrent value (2 +) |
| UU      | Building design and construction                   | Regular knowledge exchange, driven mostly by one partner, who desired to be involved earlier and more fully in the bidding process | Aligned to design and build an efficient project infrastructure                                  | Lack of informal safe guards (early involvement)<br>Formal safeguards (co-investment in each recurrent project)                        | Strong, with potential for recurrent value (1 +) |



capabilities, and the culture of another organization was very high in the beginning of the relationship and diminished once it was absorbed and spread among multiple employees of the partner organization. Thus, mutual exchange and formal processes were established at the start of the relationship cycle or project, while over time it became demand driven and informal. Organizational openness and transparency and the protection of intellectual property rights played an important role in facilitating knowledge transfer.

However, only three relationships were able to create value during at least one project, one relationship was able to create value only in a specific geographical area, and two relationships did not create value and were terminated after the initial efforts. We ascertained value creation in the following ways. During the development of a waste-to-energy project, value is created as soon as the project receives planning permission from local authorities. For example, because of the relationships with VV and UU, Covanta was able to receive planning permission and preliminary accreditation for government support for its 95 MW renewable energy plant. In addition, VV and Covanta successfully started to market an eco-park to other environmental technology companies focused on reprocessing and remanufacturing waste materials. Covanta benefited from the relationship with XX during three projects, which received planning permissions, creating recurrent value.

However, complementary resources and consistent knowledge sharing supported by an agreement about intellectual property were not enough to generate value in relationships with YY and WW. The goal of the relationship with YY was to further develop and commercialize a gasification technology with the technology provider YY. However, after a three-year project the collaborative effort was terminated due to lack of economic feasibility and commercial value. In this case, value would have been created once a co-developed technology became commercial. Covanta and WW's relationship was another case where complementary resources and established knowledge-sharing routines did not lead to value creation. Finally, despite the availability of complementary resources and existing knowledge sharing, Covanta terminated its 30-year relationship with ZZ when Covanta decided the two were no longer competitively aligned for expansion into regions beyond North America. Thus, we suggest the following formal proposition:

**P1.** . Complementary resources and knowledge sharing are necessary, but not sufficient for value creation from inter-firm relationships.

## 5.2. Competitive alignment

By examining the connection between complementary resources, knowledge sharing, and the relationship outcomes further, the data show that complementary resources and knowledge sharing were more likely to lead to value creation if there was competitive alignment in the relationship. Competitive alignment with external partners was critical for value creation during the initial as well as recurrent engagements during long-term relationships. Competitive alignment occurs when external business partners, engaged in an inter-firm relationship, share consistent or congruent activities and goals leading to business advantage. Examples of consistency or congruence between the activities of partners for gaining business advantage include product or services offerings, timing of technology commercialization, market opportunities and positioning, financial expectations, and co-investment. Competitive alignment stimulates partners to engage simultaneously in value creation and cost minimization behaviour by focusing on innovation, responsiveness, and efficiency. Without this alignment partners lose competitive tension and may develop ulterior motives that can complicate the integration of complementary resources and common value creation. While a firm can choose a partner with competitive alignment, competitive alignment can be affected by changes in the external environment. Thus, it is important for firms to create appropriate control mechanisms to sustain competitive alignment.

The evidence from all six relationships demonstrates a link between competitive alignment and value creation. The three successful relationships show that Covanta achieved value creation and was likely to re-engage with these partners with whom competitive alignment was present. All three companies agreed on the parameters of competitiveness and offered flexibility to accommodate the expectations of regulatory authorities. In addition, Covanta was able to establish relationships with UU, XX, and VV where quality improvements did not jeopardize the competitiveness of the joint projects. Covanta and UU created value during one project and planned to extend their relationship to other projects. Covanta and XX created value while obtaining planning permission for three projects and planned to continue its relationship in the future. Covanta and VV developed competitive alignment by having a common vision to develop the waste resource capabilities in the United Kingdom, willing to invest a substantial amount of capital in a joint venture and stay competitive throughout a long-term relationship.

In contrast, three failed relationships illustrate how the lack of competitive alignment hinders value creation. Covanta and YY did not complete their project because there was a competitive misalignment between these two companies. Their expectations regarding timing and cost of commercialization of the technology diverged. The complementary resources of YY, in the form of new technology, were very early in their development stage and required more time to become mature and cost competitive. Due to this mismatch, complementary resources and knowledge sharing only led to a demonstration project. The second relationship that did not lead to value creation (with utility/site provider WW) also lacked competitive alignment. The differences in the expectations for return on investment caused the competitive misalignment in this relationship. The complementary resources as well as open and regular knowledge sharing did not lead to a winning bid.

The relationship with ZZ demonstrates the change in competitive alignment affecting the ability of complementary resources and knowledge sharing to generate value. The value was generated as long as Covanta's goals were competitively aligned with the goals of ZZ. The relationship started with both firms having the same goal. As soon as Covanta decided to expand to the European market, it realized that the relationship with ZZ lost its competitive alignment, which led to Covanta terminating their relationship with ZZ.

By examining the evidence of the connection between complementary resources, knowledge sharing, competitive alignment, and

the outcomes of the relationships, we conclude that partners with complementary assets and knowledge sharing are likely to generate value if their goals are competitively aligned. Thus, the second proposition of this study posits that partners possessing competitive alignment generate more value from their complementary resources and knowledge sharing during inter-firm relationships. Based on these findings, we formally state our second proposition in the following way:

**P2.** The greater the competitive alignment of partners, the greater the potential to generate value from complementary resources and knowledge sharing during inter-firm relationships.

### 5.3. Formal control mechanisms

The relational view emphasizes the importance of informal safeguards such as openness, transparency, goodwill, and trust embedded in social relations as an important safeguard against opportunism (Kruse et al., 2022; Dyer et al., 2018; Ring and Van de Ven, 1992; Dyer, 1997; Verdecho et al., 2012). These relational agreements are known to improve confidence, facilitate personal interactions, and enhance execution of project tasks (Ariño and de la Torre, 1998; Carson et al., 2006). They also reduce the need for complex contracts resulting in lower transaction costs (Sakka et al., 2016; Ring and Van De Ven (1994)).

This study's evidence indicates that while informal safeguards are important, competitive alignment sealed with formal control mechanisms is especially effective for value creation. Competitive alignment can be enabled by a common goal to win a project, early involvement of a partner in the project, and trust development. However, these informal safeguards were not enough to reinforce competitive alignment for value creation. The relationships with low potential for value creation and re-engagement lacked formal control mechanisms. Thus, formal control mechanisms, for example contractual agreements and financial obligations, that align competitive expectations between partners, may increase the propensity for recurrent value creation. In all cases with several past or future engagements we observe competitive alignment being supported by formal safeguards in the form of contractual agreements and financial agreements. The possible explanation for this is that trust may establish transparency and a collaborative atmosphere reducing opportunistic behaviour, but it cannot necessarily ensure value creation. For example, in the case of the relationship between Covanta and VV, competitive alignment was developed through a durable safeguard initiated by VV. VV initially developed a property and obtained planning consent from the local authorities for multiple waste treatment projects in an eco-park. This important complementary resource allowed them to develop a relationship with Covanta including two long-term projects. The relationship between Covanta and UU also exhibits a positive association between competitive alignment and repeated engagement. It shows that competitive alignment can be stimulated through formal safeguards such as co-investments in the initial and subsequent projects. Thus, the formal agreements incentivizing partners to remain competitively aligned on a continuous basis enable long-term relationships. In formal terms we propose,

**P3.** The competitive alignment supported by formal control mechanisms is more likely to lead to long-term value creation from complementary resources and knowledge sharing during inter-firm relationships than the competitive alignment supported by informal control mechanisms.

## 6. Conclusion

Examining the relationship management process of inter-organizational partnerships, including identification and selection of a partner, contract negotiation, and project execution, this paper develops three propositions that help firms assess their relationships regarding initial and recurrent value creation potential. This study's insights initially explore the relationship between complementarities, knowledge sharing, and value creation. The analysis shows that companies engaged in inter-firm relations select partners with different complementary resources ranging from static resources such as land and capital to dynamic capabilities such as innovation capability. They also regularly engage in knowledge sharing. However, while the ability of a partner to offer complementary resources and engage in knowledge sharing is important, it is not sufficient to create value that can be translated into the economic gain. In fact, while in all six cases presented in this paper complementary resources and knowledge sharing existed, the value creation was limited in some cases. Thus, no clear connection between these two factors and value creation could be determined without considering other aspects.

Thus, the concept of competitive alignment is introduced to differentiate between the successful and less successful relationships. Competitive alignment represents a mechanism that stimulates partners to engage in value creation behaviour such as quality and cost improvements as well as controls for opportunistic behaviour without jeopardizing trust. This research shows that competitive alignment is an important moderator of the relationships between complementarities and knowledge sharing and the outcomes of the relationships. In all six cases, competitive alignment played an important role in either facilitating or preventing value creation. Our study also shows that the success of the initial engagement develops confidence in the relationship and motivates partners to engage in the subsequent projects allowing for long-term value creation. However, the success is contingent upon the existence of competitive alignment. Relationships that possess competitive alignment supported by formal control mechanisms are more likely to achieve not only initial but also recurrent value resulting from long-term relationships. Based on our propositions, managers can learn how to assess each engagement with external partners in terms of complementary resources, knowledge-sharing aptitude, and competitive alignment to create long-term value. In addition, the importance of exploiting formal control mechanisms to incentivize the competitive alignment behaviour on a continuous basis is depicted. Thus, firms need to rely on governance mechanisms that create value and control for opportunistic behaviour without jeopardizing trust.

This research has some limitations and opportunities for extension. First, the focus on one firm and six of its partners raises concerns

regarding the generalizability of the results. Nevertheless, the richness of the data and the complexity of the relationships offer a useful opportunity to explore how to be most effective at relationship management to create initial and long-term, recurrent project value. Second, this research focuses on the waste-to-energy industry and may raise questions about its transferability of the results. However, the lessons learned from this dynamic and complex industry where partnering is critical for success offer insight for other industries facing uncertain environments where the strength of one's teams can offer competitive advantage and value creation. Finally, future empirical research can test the findings presented here and address the limitations by drawing on more comprehensive datasets. Referring to Dyer et al. (2018), an extension of this current research could be an investigation of the effects of inter-firm relations over time and the dynamic nature of knowledge sharing and complementary resources, as well as using competitive alignment as a moderator and formal control mechanisms for stimulating sustained long-term value creation.

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